



David Matzek

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CONTACT

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LANGUAGES

German

Native



English

C1 – Oxford, UK & Chicago, USA



Spanish

B1 – Monterrey, Mexico



KEY COMPETENCIES

- > AI Agent Architecture & Automation
- > Data Governance & Ownership
- > Data & Interface Standardization
- > Model-Based Systems Engineering
- > Stakeholder Management
- > Strategic Decision Making

SENIOR DATA & SYSTEMS ARCHITECT

AI-agent automation for software-defined data platforms

Senior data & systems architect (8+ years) who designs multi-agent AI systems that automate governance, CI/CD and release work — removing manual overhead and hard access barriers while keeping long-term architecture clean and maintainable.

PROFESSIONAL HIGHLIGHTS.

- > Built a 7-agent AI system (~15 reusable skills) automating governance work equal to ~3 FTEs (~€350k/year) for 45+ contributors across 180+ applications.
- > Cut daily operating effort from 20 h to 35 min for a 240-consumer data model after a staffing and budget cut.
- > Delivered a metadata self-service capability in 20 workdays, closing a request open for 6 years.
- > Owner of BMW's VSS vehicle data model; active COVESA contributor; 3 patent specifications.

PROFESSIONAL POSITIONS.

01/2025 –
today

Data Steward & Data Model Owner | BMW Group

Vehicle Data Assets | EL-430

- > Own BMW's VSS vehicle-data model (Vehicle Shadow), consumed by 180+ backend applications.
- > Built 7 AI agents + a self-service "VSS Explorer" UI automating PR triage, CI-fixing, EU Data Act classification and releases (~3 FTEs saved).
- > Cut daily effort 20 h → 35 min; driving the .vspec → GraphQL (S2DM) model migration.

01/2023 –
12/2024

Lead Data Designer | BMW Group

Vehicle Data Infrastructure | EE-340

- > Established an early-phase data-design discipline across a 60+ ECU board net.
- > Set E/E (physical bus) requirements so needed data reaches in-car collectors by design.

11/2021 –
12/2022

E/E Data Architect | BMW Group

Data Strategy, Connected Vehicle | EE-313

- > Middleware-based (ADAPT) standardization reducing model fragmentation across platforms.
- > Variant management and test automation for cross-derivative data.

04/2017 –
11/2021

Specialist Function Development | BMW Group

Intelligent Personal Assistant | DE-55

- > Functional owner of intelligent comfort functions; 4 in-car functions released to series.
- > End-to-end system design over the V-model; generated 2 patent specifications.

SOFTWARE & TOOLS

VS Code & GitHub

Development, versioning, automation



GitHub Copilot & MCP

Agentic workflows, token optimization



Atlassian JIRA & Confluence

Process & project lead



ALM Octane

Functional owner, SW quality



PowerPoint

Executive-level storytelling



QUALIFICATIONS

- > Internal first aider
- > Internal driving licenses

INTERESTS

- > Digitalization & entrepreneurship
- > Endurance sports
- > Personal development
- > Cooking & coffee

SELECTED CASE — AI-AGENT AUTOMATION

01/2026 –
today

VSS Data-Model Governance, Automated

BMW Group | Vehicle Shadow

- > 7 purpose-built agents (~15 skills) orchestrate PR triage, CI-fixing, auto-rebase, EU Data Act classification, changelog and version bumps, and release-board sorting.
- > “VSS Explorer” self-service UI lets non-technical contributors request signals without git/PR skills; a backend pipeline auto-generates compliant PRs.
- > Cost-aware LLM tooling: description rewriter benchmarked across 3 model tiers for best cost/quality.
- > Impact: ~3 FTEs (~€350k/year) of governance work removed for 45+ users at a weekly release cadence.

PATENT SPECIFICATIONS.

WO2020083962A1

Apparatus and control unit for automating a state change of a window pane of a vehicle.

WO2020011655A1

Method and system for detecting data in vehicles.

DE102018209967A1

Method for automatically configuring a function in a vehicle.

EDUCATION

10/2018 –
10/2021

M.Sc. Systems Engineering

University of Applied Sciences Munich

- > In parallel to professional position at BMW.
- > Thesis: influence of frontloading on software-development efficiency.

10/2013 –
01/2017

B.Eng. Mechatronics

University of Applied Sciences Esslingen

- > Focus on automotive electronics; BMW SpeedUp future-talent program.

08/2015 –
01/2016

Exchange Semester

Tecnológico de Monterrey, Mexico

- > Cross-cultural experience; thesis on external perception of autonomous vehicles.

09/2010 –
07/2013

Apprenticeship Mechatronics (IHK)

BMW Group, Munich

- > In-house project lead with €4M scope.